



Backend Engineer – Assignment

Build a scalable healthcare appointment platform using both **Spring Boot** and **Python** services.



Timeline

- ⌚ Duration: 24 to 48 Hours
 - ⌚ Expected Effort: 6 to 12 Hours
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What You Need to Build

Build a web based healthcare appointment management platform with:

- ☕ A Spring Boot backend service
- 🐍 A Python worker/service
- 📡 Event-driven communication using Kafka/RabbitMQ
- 🗄 Database integration
- 🔒 Authentication & authorization

Think of it as a simplified production-style backend system.



Example Workflow

1. 📅 User books appointment via Spring Boot API
 2. 📡 Spring Boot publishes event to Kafka/RabbitMQ
 3. 🐍 Python service consumes event
 4. 🔔 Notification sent
 5. ✅ Processing status gets updated
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Suggested Tech Stack

Layer	Options
Backend Core	SpringBoot
Worker Service	Python

Database	H2/Postgres/Any Other Database
Messaging	Kafka/RabbitMQ
API Docs	Swagger
Containerization	Docker
Frontend	ViteJs/NextJS
Cloud	AWS/GCP/Azure

✓ Core Requirements

1. ⚙️ APIs & Features

Build APIs for:

- 👤 Register/Login User
 - 📅 Create Appointment
 - ❌ Cancel Appointment
 - 📖 Fetch User Appointments
 - ⌚ Fetch Available Slots
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2. 🗄️ Database & Business Logic

- ✓ Proper schema design
 - ⚡ Prevent duplicate booking
 - ⚡ Handle concurrent requests
 - 📝 Maintain appointment logs/history
 - ⌚ Store timestamps and status updates
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3. 🔄 Event-driven Communication

- 📡 Publish events from Spring Boot
 - 🐍 Consume/process events in Python service
 - 🔄 Handle event processing properly
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4. 🔑 Security & Architecture

- 🔑 JWT-based authentication

-  Proper validation and exception handling
 -  Clean architecture and modular code structure
 -  API documentation
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5. UI Requirement

-  Simple frontend/UI to demonstrate backend workflows and APIs
-  User Registration/Login interface
-  Appointment booking and management screens
-  View appointment history and available slots
-  Show event/processing status visually on UI

Example

-  "Fetching available slots..."
 -  "Booking appointment..."
 -  "Appointment booked successfully"
 -  "Processing notification event..."
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Deliverables

-  Github Repository - Backend and Frontend
 -  API Documentation & Swagger Link
 -  Database Schema
 -  README with setup instructions
 -  Demo Video / Screenshots
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Evaluation Criteria

Must Have

-  Working Spring Boot + Python services
-  Service-to-service communication
-  Database integration
-  Kafka/Event-driven workflow
-  Clean architecture and code quality
-  Smooth appointment workflow

● Bonus Points

-  Dockerized deployment
-  Better performance optimization
-  Production-ready engineering practices